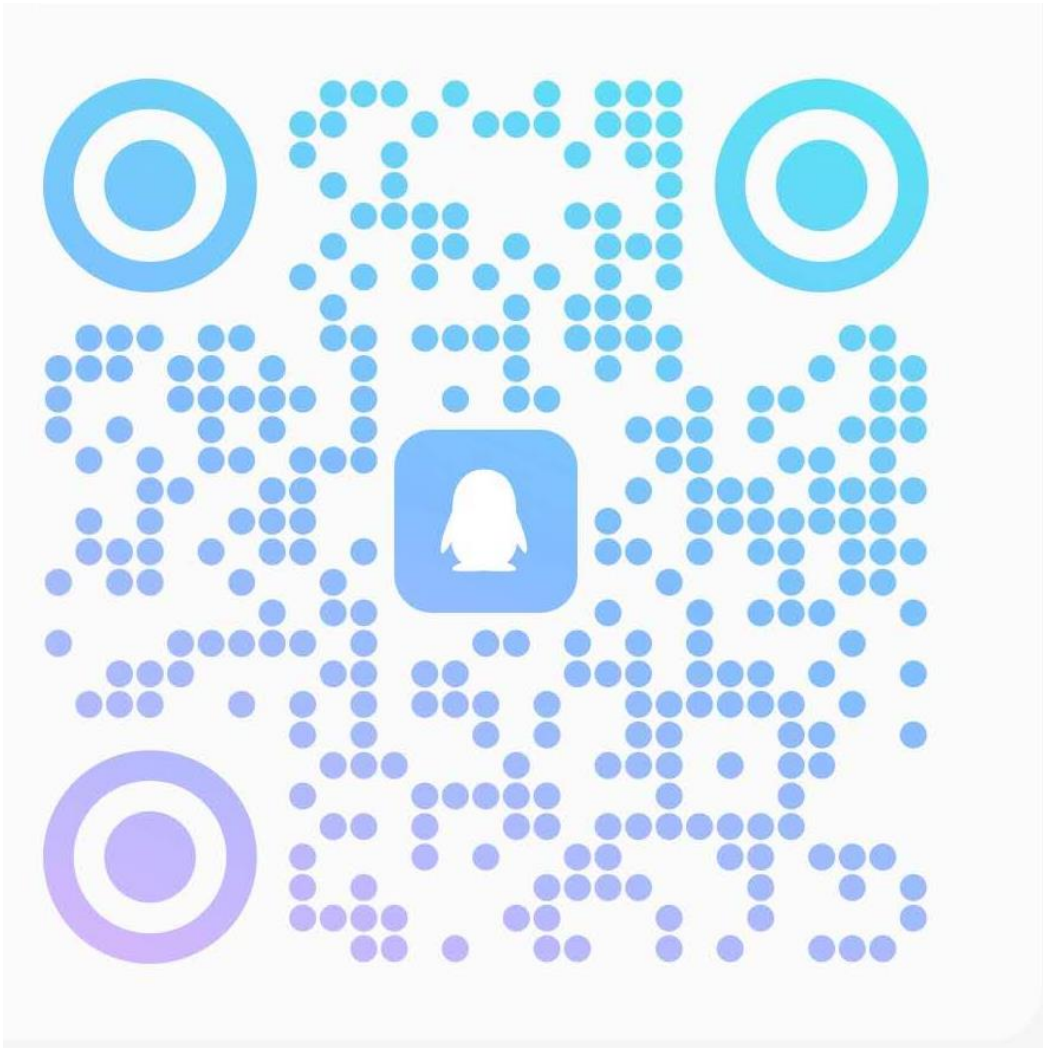




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Chapter 2

Descriptive Statistics:

Tabular and Graphical Methods

In Chapter 1 we saw that although we can sometimes take a census of an entire population, we often must randomly select a sample from a population.

When we have taken a census or a sample, we typically wish to describe the observed data set. In particular, we describe a sample in order to make inferences about the sampled population.

In this chapter we begin to study descriptive statistics, which is the science of describing the important characteristics of a data set.

The techniques of descriptive statistics include tabular and graphical methods, which are discussed in this chapter, and numerical methods, which are discussed in Chapter 3. We will see that, in practice, the methods of this chapter and the methods of Chapter 3 are used together to describe data. We will also see that the methods used to describe quantitative data differ somewhat from the methods used to describe qualitative data.

Chapter Outline

- 2.1 Graphically Summarizing Qualitative Data
- 2.2 Graphically Summarizing Quantitative Data
- 2.3 Dot Plots
- 2.4 Stem-and-Leaf Displays

2.1 Graphically summarizing qualitative data

- With qualitative data, names identify the different categories.
- This data can be summarized using a frequency distribution.
- **Frequency distribution:** A table that summarizes the number (or frequency, e.g., how many times) of items in each of several non-overlapping classes.

Example 2.1 Describing Pizza Preferences

- A business entrepreneur plans to open a pizza restaurant in a college town.
- The entrepreneur wishes to study the pizza preferences of the college students in town.
- In order to do this, the entrepreneur selects a random sample of 50 students enrolled in the local college and asks each sampled student to name his or her favourite among the six pizza places in town.
- The survey results are given in Table 2.1.

Example 2.1 - Continue

TABLE 2.1 Pizza Preferences of 50 College Students 

Little Caesars	Papa John's	Bruno's	Papa John's	Domino's
Papa John's	Will's Uptown	Papa John's	Pizza Hut	Little Caesars
Pizza Hut	Little Caesars	Will's Uptown	Little Caesars	Bruno's
Papa John's	Bruno's	Papa John's	Will's Uptown	Papa John's
Bruno's	Papa John's	Little Caesars	Papa John's	Little Caesars
Papa John's	Little Caesars	Bruno's	Will's Uptown	Papa John's
Will's Uptown	Papa John's	Will's Uptown	Bruno's	Papa John's
Papa John's	Domino's	Papa John's	Pizza Hut	Will's Uptown
Will's Uptown	Bruno's	Pizza Hut	Papa John's	Papa John's
Little Caesars	Papa John's	Little Caesars	Papa John's	Bruno's

The raw data. What information you can get from these data?

Continue: The resulting **frequency distribution**

TABLE 2.2 A Frequency Distribution of Pizza Preferences


Restaurant	Frequency
Bruno's	8
Domino's	2
Little Caesars	9
Papa John's	19
Pizza Hut	4
Will's Uptown	8
	50

See Excel file

The purpose of the frequency distribution is to make the data easier to understand. Certainly, looking at the frequency distribution in Table 2.2 is more informative than looking at the raw data in Table 2.1.

We see that Papa John's is the most popular restaurant, and that Papa John's is roughly twice as popular as each of the next three runners up—Bruno's, Little Caesars, and Will's. Finally, Pizza Hut and Domino's are the least preferred restaurants.

Relative frequency

- The **relative frequency** of a class is the proportion or fraction of data that is contained in that class.
 - Calculated by dividing the class frequency by the total number of data values.
 - Relative frequency may be expressed as either a decimal or percent
 - A **relative frequency distribution** is a list of all the data classes and their associated relative frequencies.

Relative frequency and percent frequency

- Relative frequency summarizes the proportion of items in each class.
- For each class, **divide the frequency** of the class by the total number of observations.
- Multiply **times 100** to obtain the percent frequency

$$\text{Relative frequency of a class} = \frac{\text{frequency of the class}}{\text{The size of the sample}}$$

TABLE 2.3 Relative Frequency and Percent Frequency Distributions for the Pizza Preference Data


Restaurant	Relative Frequency	Percent Frequency
Bruno's	$8/50 = .16$	16%
Domino's	.04	4%
Little Caesars	.18	18%
Papa John's	.38	38%
Pizza Hut	.08	8%
Will's Uptown	.16	16%
	<u>1.0</u>	<u>100%</u>

A relative frequency distribution is a table that lists the relative frequency for each class, and a percent frequency distribution lists the percent frequency for each class.

Table 2.3 gives a relative frequency distribution and a percent frequency distribution of the pizza preference data.

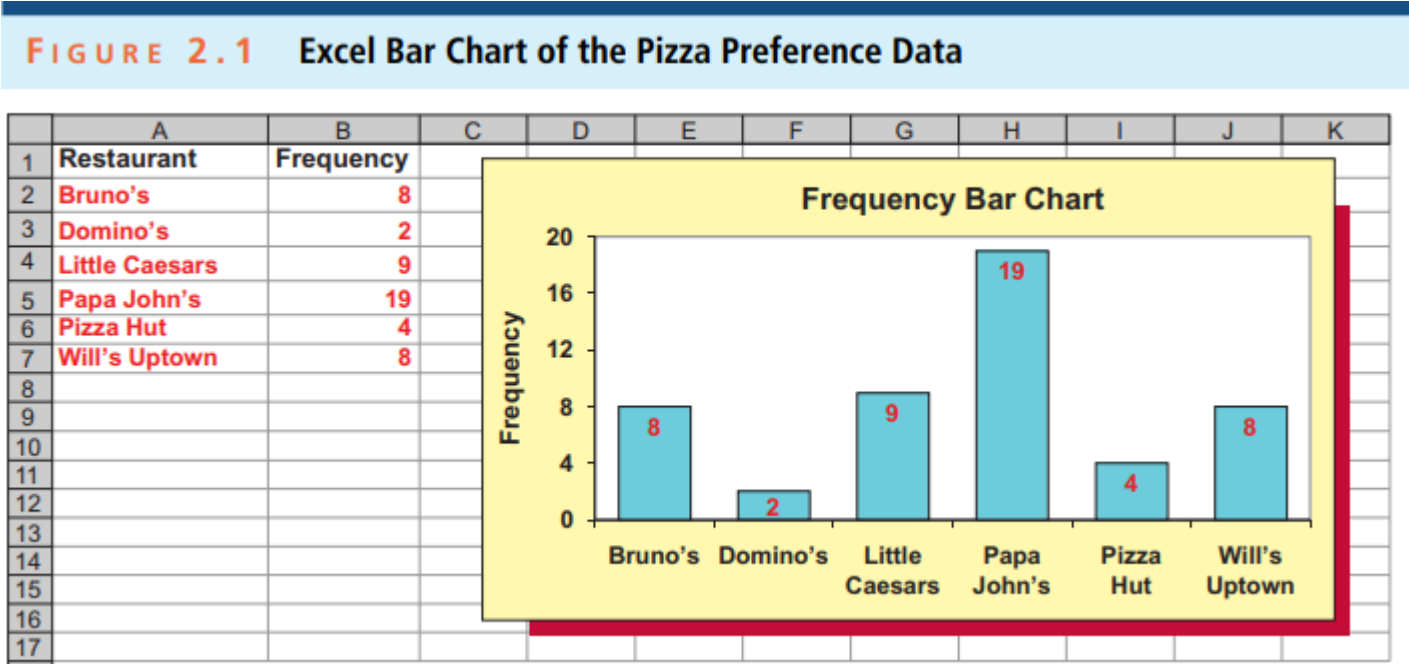
The relative frequency for Papa John's pizza is $19 / 50 = 0.38$. this means 38% of the sampled students preferred Papa John's pizza.

The sum of the relative frequencies in the relative frequency distribution equals 1.0, and the sum of the percent frequencies in the percent frequency distribution equals 100%.

Bar charts and Pie charts

- **Bar chart:** A vertical or horizontal rectangle represents the frequency for each category
 - Height can be frequency, relative frequency, or percent frequency
- **Pie chart:** A circle divided into slices where the size of each slice represents its relative frequency or percent frequency.

Excel Bar Chart of Pizza Preference



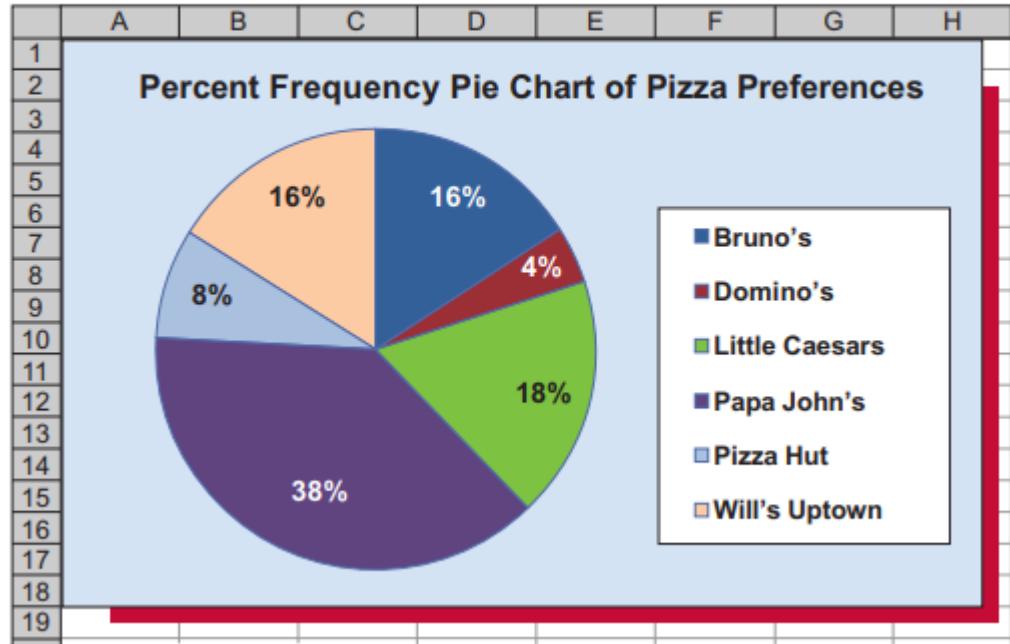
- A bar chart is a graphic that depicts a frequency, relative frequency, or percent frequency distribution. For example, Figure 2.1 gives an Excel bar chart of the pizza preference data. On the horizontal axis we have placed a label for each class (restaurant), while the vertical axis measures frequencies.

- Because the height of each bar is a frequency, we refer to Figure 2.1 as a frequency bar chart.
- The bar chart in Figure 2.1 clearly illustrates that, for example, Papa John's pizza is preferred by more sampled students than any other restaurant and Domino's pizza is least preferred by the sampled students.

See Excel file

Excel Pie Chart of Pizza Preference

FIGURE 2.3 Excel Pie Chart of the Pizza Preference Data



A pie chart is a graphic that can be used to depict a frequency distribution. When constructing a pie chart, we first draw a circle to represent the entire data set. We then divide the circle into sectors or “pie slices” based on the relative frequencies of the classes.

For example, remembering that a circle consists of 360 degrees, Bruno's Pizza (which has relative frequency 0.16) is assigned a pie slice that consists of $0.16 \times 360 = 57.6$ degrees.

The resulting pie chart (constructed using Excel) is shown in Figure 2.3. Here we have labelled the pie slices using the **percent frequencies**. The pie slices can also be labelled using **frequencies** or **relative frequencies**.

See Excel file

2.2 Graphically summarizing quantitative data

- Often need to summarize and describe the shape of the distribution of a population or sample of measurements.
- One way is to group the measurements into classes of a frequency distribution and then displaying the data in the form of a histogram.

Example 2.2 The e-Billing Case

It is hoped that a new billing system will substantially reduce the amount of time it takes customers to make payments.

In order to assess the system's effectiveness, a random sample of 65 invoices was selected from the 7,823 invoices processed during the first three months of the new system's operation.

The payment times for the 65 sample invoices are manually determined and are given in Table 2.4.

The objective is to see if this sample can be used to establish that the new billing system substantially reduces payment times.

Example 2.2 – continue: The e-Billing Case

TABLE 2.4 A Sample of Payment Times (in Days) for 65 Randomly Selected Invoices 

22	29	16	15	18	17	12	13	17	16	15
19	17	10	21	15	14	17	18	12	20	14
16	15	16	20	22	14	25	19	23	15	19
18	23	22	16	16	19	13	18	24	24	26
13	18	17	15	24	15	17	14	18	17	21
16	21	25	19	20	27	16	17	16	21	

- The shortest payment time is 10 days and that the longest payment time is 29 days.
- Beyond that, it is pretty difficult to interpret the data in any meaningful way.
- To better understand the sample of 65 payment times, a frequency distribution of the data can be formed and this can graph the distribution by constructing a histogram.
- The frequency distribution will divide the payment times into classes and will tell us how many of the payment times are in each class.

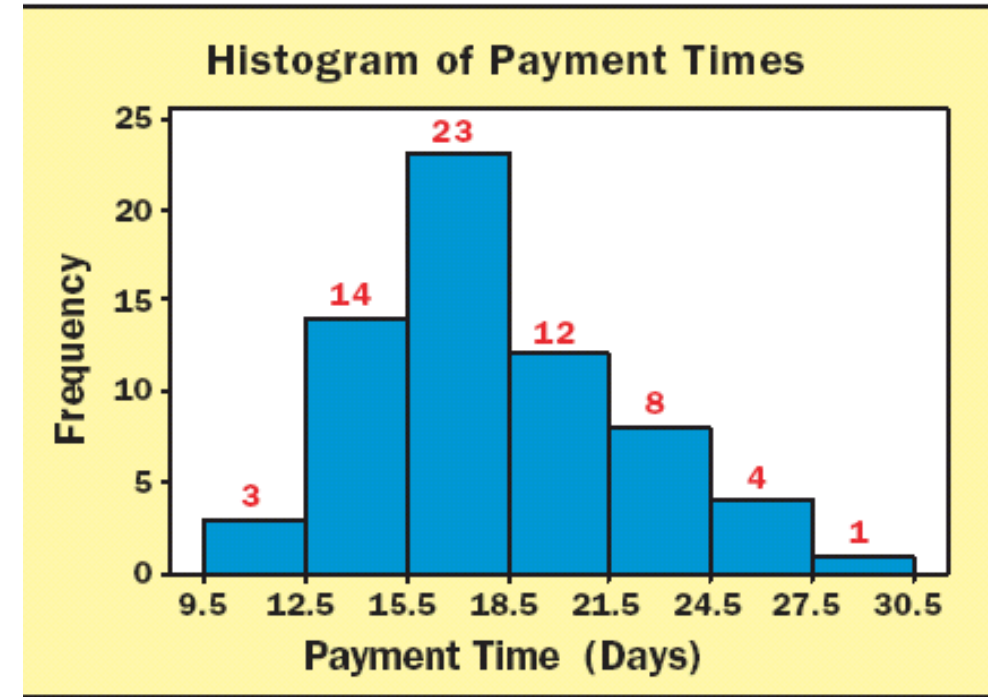
Frequency Distribution and Histogram

A **frequency distribution** is a list of data classes with the count or “frequency” of values that belong to each **class**

- “Classify and count”
- The frequency distribution is a table.

Show the frequency distribution in a **histogram**

- The histogram is a picture of the frequency distribution.



See Excel file

Steps in Constructing a Frequency Distribution

Step 1: Find the number of classes,

Step 2: Find the class length,

Step 3: Form non-overlapping classes of equal length,

Step 4: Tally and count the number of measurements in each class,

Step 5: Graph the histogram.

Example 2.2 – The e-Billing Case

TABLE 2.4 A Sample of Payment Times (in Days) for 65 Randomly Selected Invoices 

22	29	16	15	18	17	12	13	17	16	15
19	17	10	21	15	14	17	18	12	20	14
16	15	16	20	22	14	25	19	23	15	19
18	23	22	16	16	19	13	18	24	24	26
13	18	17	15	24	15	17	14	18	17	21
16	21	25	19	20	27	16	17	16	21	

The largest is 29 days

The smallest is 10 days

Step 1: Number of classes

- Group all of the n data into K number of classes
- K is the smallest whole number for which $2^K \geq n$
- In example 2.2, $n = 65$
 - For $K = 6$, $2^6 = 64, < n$
 - For $K = 7$, $2^7 = 128, > n$
 - So use $K = 7$ classes

Step 2: Class length

- Find the length of each class as the largest measurement minus the smallest divided by the number of classes found earlier (K)
- For **example 2.2**, $(29-10) / 7 = 2.7143$
 - Because payments measured in days, round **up** to **three** days

$$\text{Approximate class length} = \frac{\text{Largest measurement} - \text{smallest measurement}}{\text{Number of classes}}$$

Step 3: Form non-overlapping classes of equal width

Class 1	10 days and less than 13 days (i.e., 10, 11, 12)
Class 2	13 days and less than 16 days
Class 3	16 days and less than 19 days
Class 4	19 days and less than 22 days
Class 5	22 days and less than 25 days
Class 6	25 days and less than 28 days
Class 7	28 days and less than 31 days (i.e., 28, 29, 30)

Step 4: Tally and count the number of measurements in each class

Class	First 4 Tally Marks	All 65 Tally Marks	Frequency
10 < 13		///	3
13 < 16		//// //// ////	14
16 < 19	//	//// //// //// //// ///	23
19 < 22	/	//// //// //	12
22 < 25	/	//// ///	8
25 < 28		////	4
28 < 31		/	1

We simply list the classes, examine the payment times in Table 2.4 one at a time, and record a tally mark corresponding to a particular class each time we encounter a measurement that falls in that class. For example, because the first four payment times in Table 2.4 are 22, 19, 16, and 18, the first four tally marks are shown above.

TABLE 2.7 Frequency Distributions of the 65 Payment Times

Class	Frequency	Relative Frequency	Percent Frequency
10 < 13	3	$3/65 = .0462$	4.62%
13 < 16	14	$14/65 = .2154$	21.54
16 < 19	23	.3538	35.38
19 < 22	12	.1846	18.46
22 < 25	8	.1231	12.31
25 < 28	4	.0615	6.15
28 < 31	1	.0154	1.54

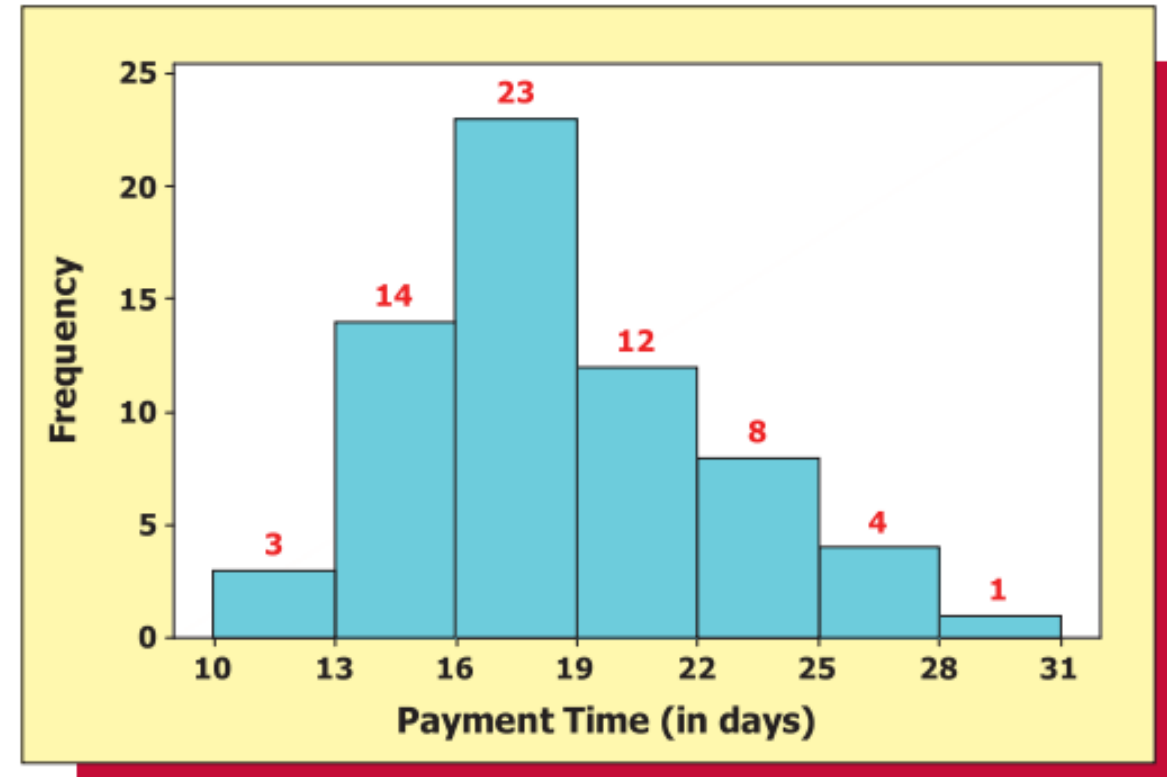
See Excel file

Step 5: Graph the histograms

The histogram can be constructed using the frequency, relative frequency, or percent frequency distribution.

To set up the histogram, we draw rectangles that correspond to the classes. The base of the rectangle corresponding to a class represents the payment times in the class. The height of the rectangle can represent the **class frequency**, **relative frequency**, or **percent frequency**.

FIGURE 2.7 A Frequency Histogram of the 65 Payment Times

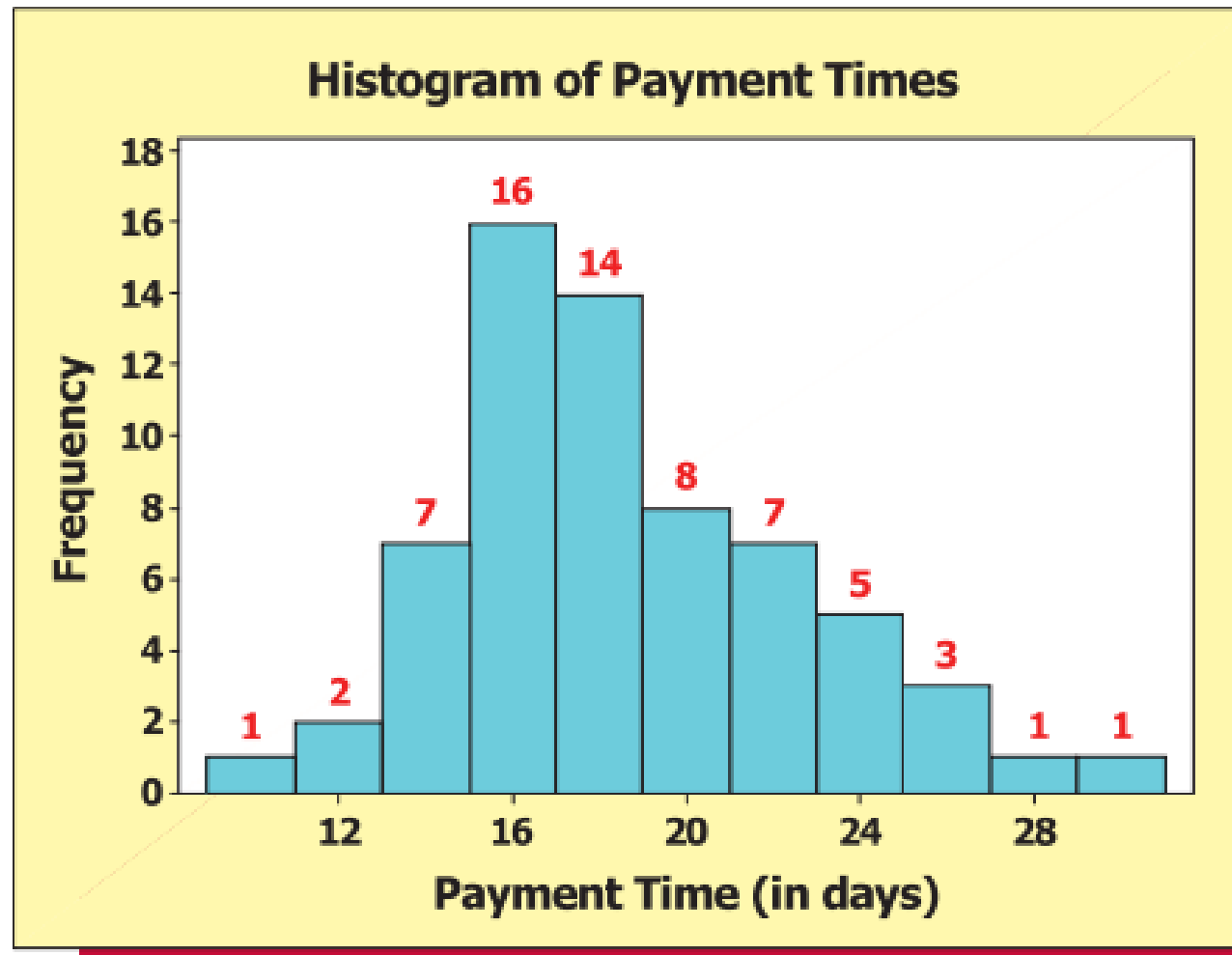


The procedure in the above box is not the only way to construct a histogram. Often, histograms are constructed more informally. For instance, we can [define 11 classes](#) and label 5 of the classes on the horizontal axis using midpoints (12, 16, 20, 24, 28). It is easy to see that the midpoints of the un-labeled classes are 10, 14, 18, 22, 26, and 30. Moreover, the boundaries of the first class are 9 and 11, the boundaries of the second class are 11 and 13, and so forth.

TABLE 2.4 A Sample of Payment Times (in Days) for 65 Randomly Selected Invoices 

22	29	16	15	18	17	12	13	17	16	15
19	17	10	21	15	14	17	18	12	20	14
16	15	16	20	22	14	25	19	23	15	19
18	23	22	16	16	19	13	18	24	24	26
13	18	17	15	24	15	17	14	18	17	21
16	21	25	19	20	27	16	17	16	21	

FIGURE 2.10 A MINITAB Frequency Histogram of the Payment Times with Automatic Classes: The Payment Time Distribution Is Skewed to the Right



Five minutes break

Please sign your attendance

Skewed Distributions

FIGURE 2.9 A Percent Frequency Histogram of the Gas Mileages: The Gas Mileage Distribution Is Symmetrical and Mound Shaped

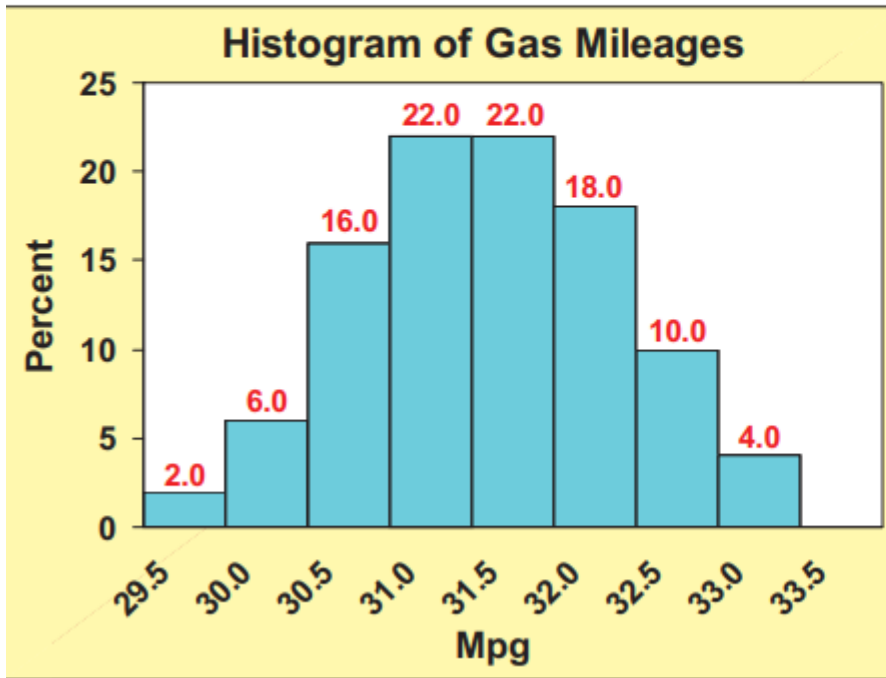
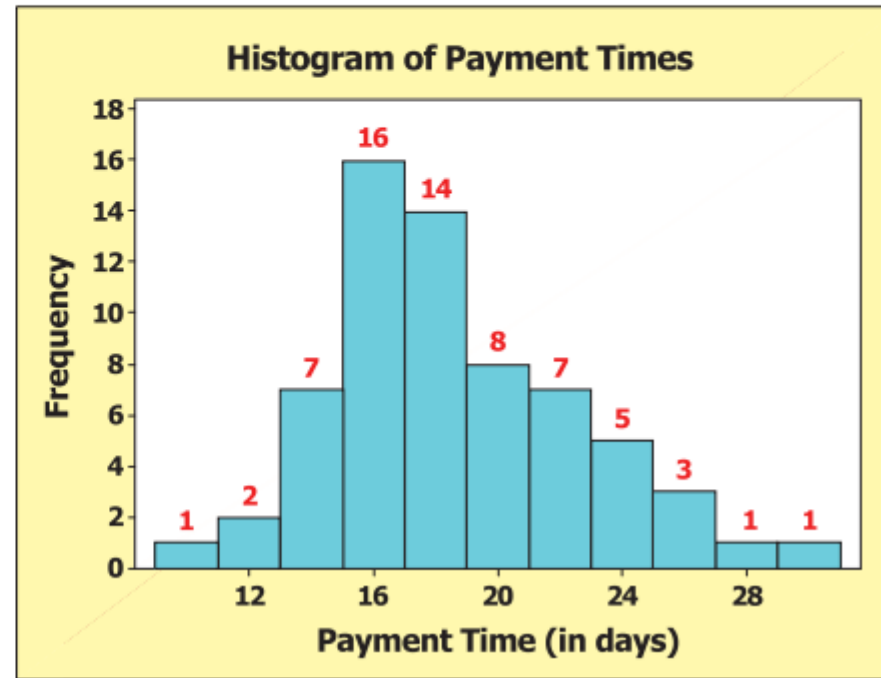


FIGURE 2.10 A MINITAB Frequency Histogram of the Payment Times with Automatic Classes: The Payment Time Distribution Is Skewed to the Right



Skewed Distributions

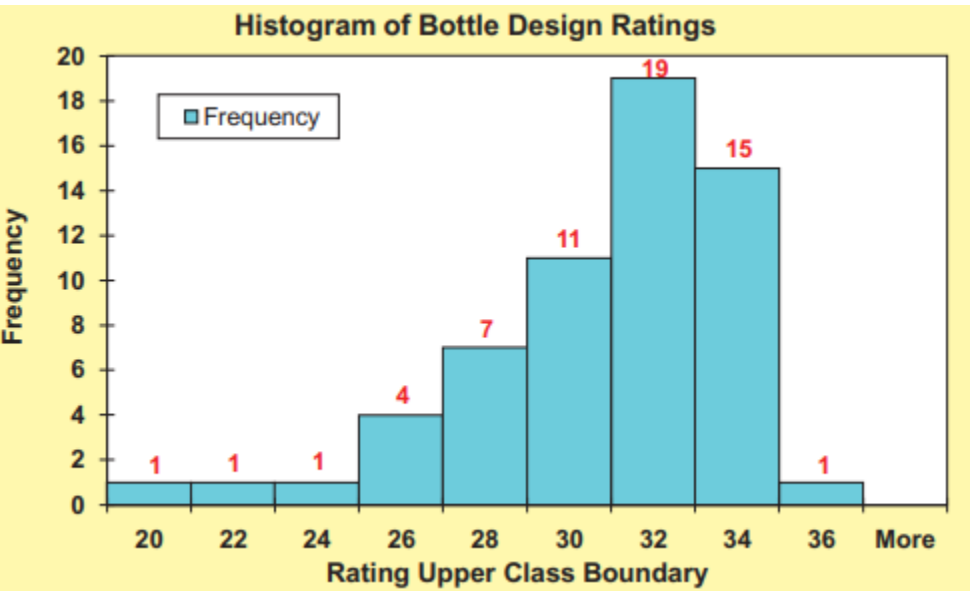


FIGURE 2.9 A Percent Frequency Histogram of the Gas Mileages: The Gas Mileage Distribution Is Symmetrical and Mound Shaped

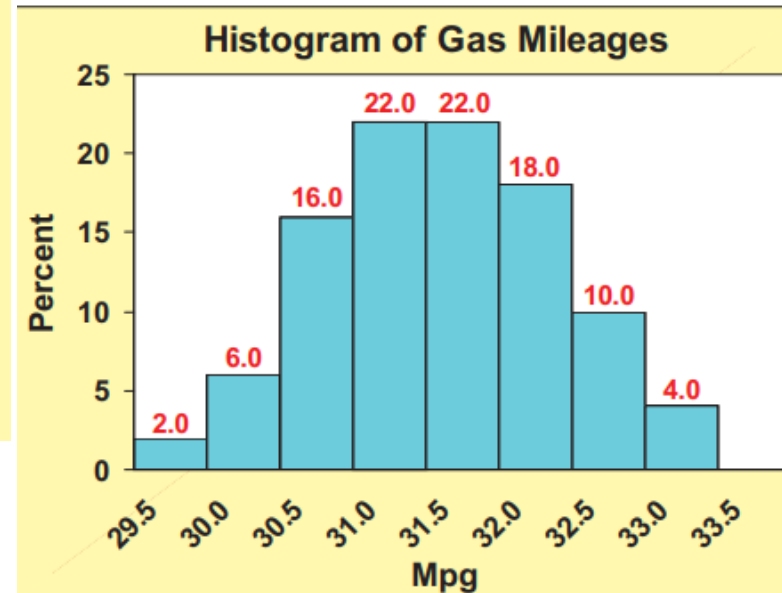
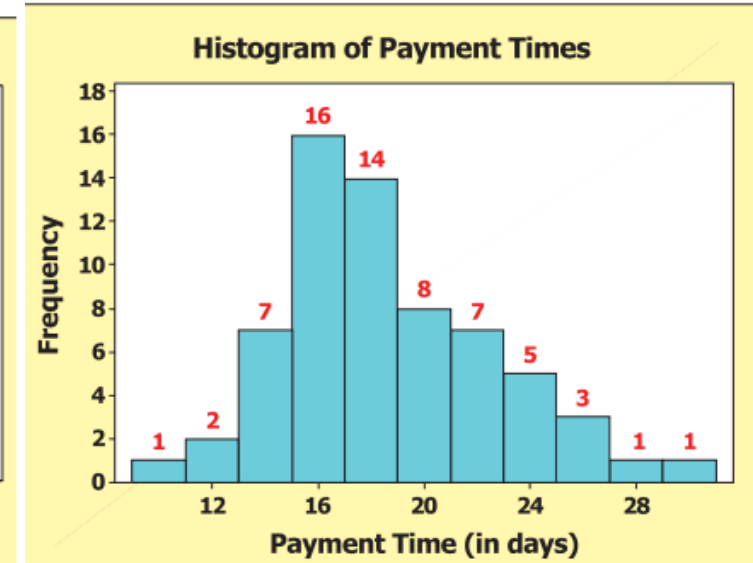


FIGURE 2.10 A MINITAB Frequency Histogram of the Payment Times with Automatic Classes: The Payment Time Distribution Is Skewed to the Right

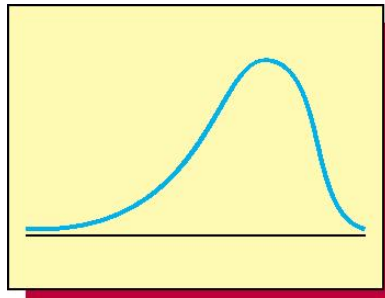


Skewness

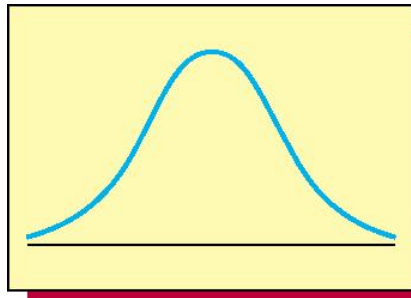
Skewed distributions are not symmetrical about their center. Rather, they are lop-sided with a longer tail on one side or the other.

- A population is distributed according to its relative frequency curve
- The skew is the side with the longer tail

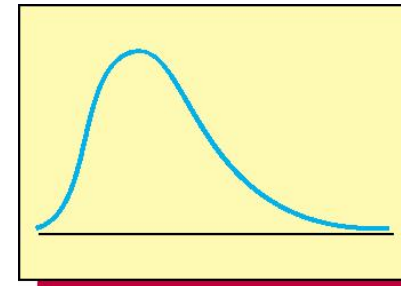
Left Skewed



Symmetric



Right Skewed



Other distributions

Many other distribution shapes are possible. For example, some distributions have two or more peaks—we will give an example of this distribution shape later in this section. It is often very useful to know the shape of a distribution.

Example 2.3 Grade Distributions

- Table 2.8 lists the scores earned on the first exam by the 40 students in a business statistics course. Figure 2.12 gives a percent frequency polygon for these exam scores.

TABLE 2.8 Exam Scores for the First Exam Given in a Statistics Class				
Class DS FirstExam				
32	63	69	85	91
45	64	69	86	92
50	64	72	87	92
56	65	76	87	93
58	66	78	88	93
60	67	81	89	94
61	67	83	90	96
61	68	83	90	98

1 (one) is between 30 – 39 scores, 2.5%

1 is between 40 – 49 scores, 2.5%

3 is between 50 – 59 scores, 7.5%

13 is between 60 – 69 scores, 32.5%

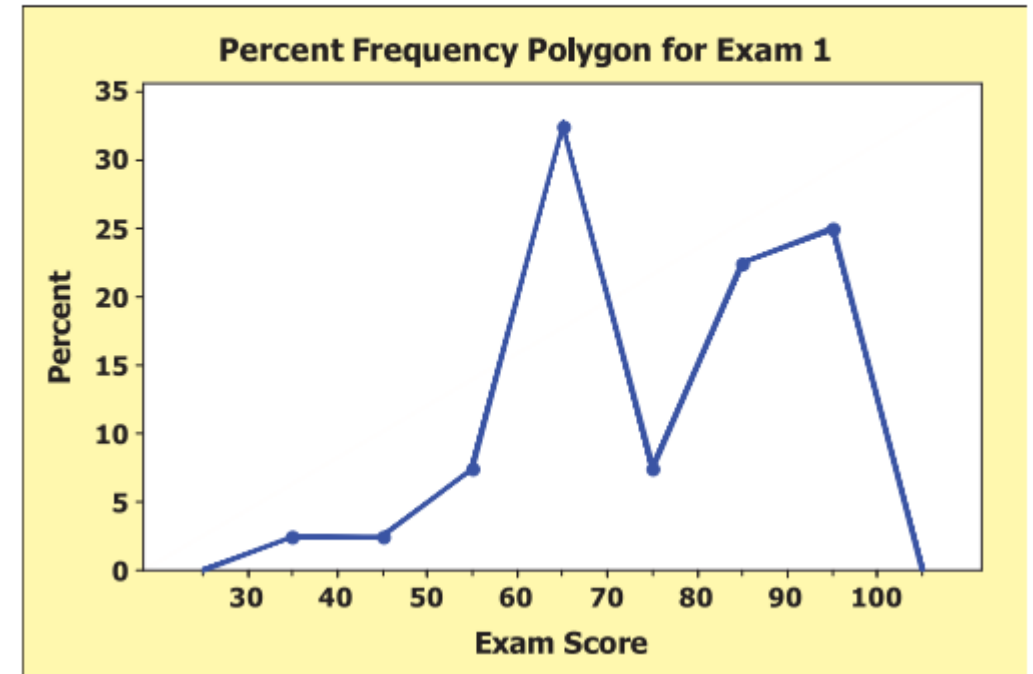
3 is between 70 – 79 scores, 7.5%

9 is between 80 – 89 scores, 22.5%

10 is between 90 – 99 scores, 25%

- Because exam scores are often reported by using 10-point grade ranges (for instance, 80 to 90 percent), we have defined the following classes: $30 < 40$, $40 < 50$, $50 < 60$, $60 < 70$, $70 < 80$, $80 < 90$, and $90 < 100$.
- The points that form the polygon have been plotted corresponding to the midpoints of the classes (35, 45, 55, 65, 75, 85, 95). Each point is plotted at a height that equals the percentage of exam scores in its class.
- For instance, because 10 of the 40 scores are at least 90 and less than 100, the plot point corresponding to the class midpoint 95 is plotted at a height of 25 percent.
- There is a concentration of scores in the 85 to 95 range and another concentration of scores around 65. In addition, the distribution of scores is somewhat skewed to the left—a few students had scores (in the 30s and 40s) that were quite a bit lower than the rest.

FIGURE 2.12 A Percent Frequency Polygon of the Exam Scores



Several Frequency Distributions

Example: Payment time - Raw data below:

TABLE 2.4 A Sample of Payment Times (in Days) for 65 Randomly Selected Invoices 

22	29	16	15	18	17	12	13	17	16	15
19	17	10	21	15	14	17	18	12	20	14
16	15	16	20	22	14	25	19	23	15	19
18	23	22	16	16	19	13	18	24	24	26
13	18	17	15	24	15	17	14	18	17	21
16	21	25	19	20	27	16	17	16	21	

Table 2.10

A Frequency Distribution, Cumulative Frequency Distribution, Cumulative Relative Frequency Distribution, and Cumulative Percent Frequency Distribution for the Payment Time Data

(1) Class	(2) Frequency	(3) Cumulative Frequency	(4) Cumulative Relative Frequency	(5) Cumulative Percent Frequency
10 < 13	3	3	$3/65 = .0462$	4.62%
13 < 16	14	17	$17/65 = .2615$	26.15
16 < 19	23	40	.6154	61.54
19 < 22	12	52	.8000	80.00
22 < 25	8	60	.9231	92.31
25 < 28	4	64	.9846	98.46
28 < 31	1	65	1.0000	100.00

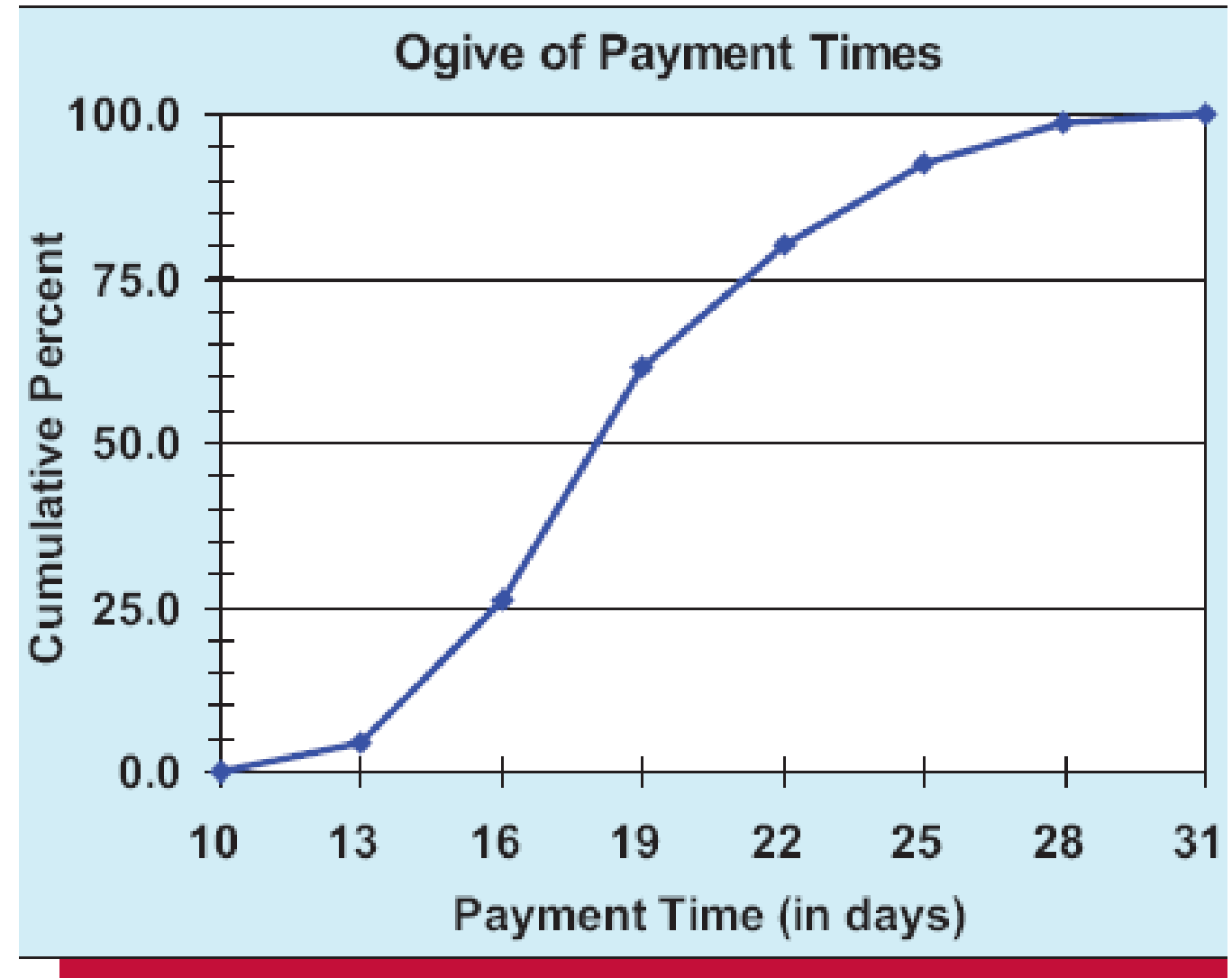
See Excel file

Cumulative Distributions

- Another way to summarize a distribution is to construct a cumulative distribution.
- To do this, use the same number of classes, class lengths, and class boundaries used for the frequency distribution.
- Rather than a count, we record the number of measurements that are less than the upper boundary of that class
 - In other words, a running total.

Ogive (pronounced “oh-jive”)

- **Ogive:** A graph of a cumulative distribution
- Plot a point above each upper class boundary at height of cumulative frequency
- Connect points with line segments
- Can also be drawn using:
 - Cumulative relative frequencies
 - Cumulative percent frequencies.



See Excel file

2.3 Dot plots

- A very simple graph that can be used to summarize a data set is called a **dot plot**.
- To make a dot plot we draw a horizontal axis that spans the range of the measurements in the data set. We then place dots above the horizontal axis to represent the measurements

Example: Distribution of scores for an exam

Raw data ⇒

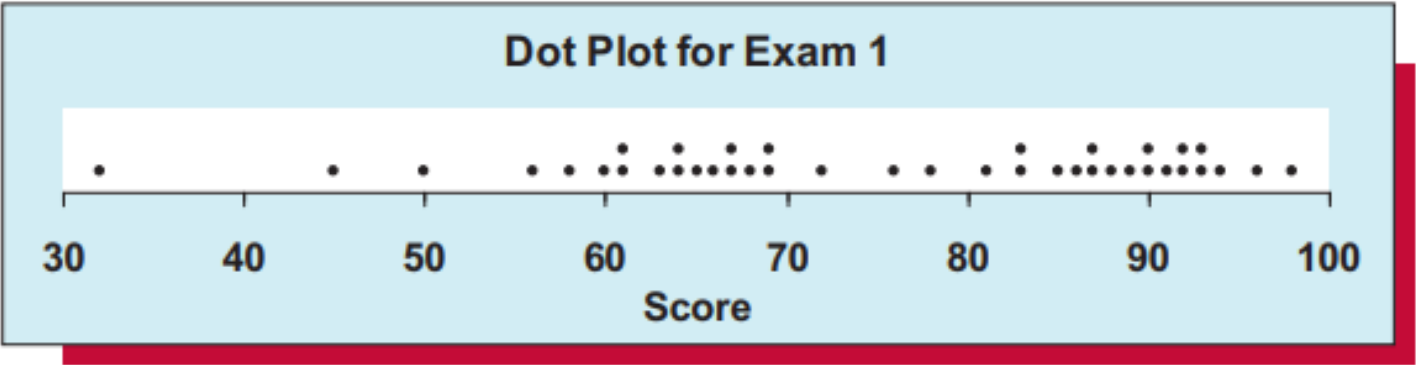
Dot data



TABLE 2.8 Exam Scores for the First Exam Given in a Statistics Class DS FirstExam				
32	63	69	85	91
45	64	69	86	92
50	64	72	87	92
56	65	76	87	93
58	66	78	88	93
60	67	81	89	94
61	67	83	90	96
61	68	83	90	98

Lowest score is 32

Dot Plot of Scores on Exam 1: Before Attendance Policy



2.4 Stem-and-leaf display

The purpose of a stem-and-leaf display is to see the overall pattern of the data, by grouping the data into classes. To see:

- the variation from class to class,
- the amount of data in each class,
- the distribution of the data within each class,
- Best for small to moderately sized data distributions.



Constructing a Stem-and-leaf Display

- Decide what units will be used for the stems and the leaves. As a general rule, choose units for the stems so that there will be somewhere **between 5 and 20 stems**.
- Place the stems in a column with the smallest stem at the top of the column and the largest stem at the bottom.
- Enter the leaf for each measurement into the row corresponding to the proper stem. The leaves should be single-digit numbers (rounded values).
- If desired, rearrange the leaves so that they are in increasing order from left to right.

Example: The Car Mileage Case

- Study of tax credit offered by the federal government for improving fuel economy
- Automaker has introduced a new model and wishes to demonstrate it qualifies for the tax credit
- Sample of 50 cars

The Car Mileage Case: The Raw Data

TABLE 2.14 A Sample of 50 Mileages for a New Midsize Model 

30.8	30.8	32.1	32.3	32.7
31.7	30.4	31.4	32.7	31.4
30.1	32.5	30.8	31.2	31.8
31.6	30.3	32.8	30.7	31.9
32.1	31.3	31.9	31.7	33.0
33.3	32.1	31.4	31.4	31.5
31.3	32.5	32.4	32.2	31.6
31.0	31.8	31.0	31.5	30.6
32.0	30.5	29.8	31.7	32.3
32.4	30.5	31.1	30.7	31.4

The largest data (longest distance travelling) is **33 miles** and the smallest is **29 miles**.

Note that the sample mileages range from 29.8 to 33.3 and we place the leading digits of these mileages — the whole numbers 29, 30, 31, 32, and 33 — in a column on the left side of a vertical line as follows:

29	
30	
31	
32	
33	

This vertical arrangement of leading digits forms the stem of the display. Next, we pass through the mileages in Table 2.14 one at a time and place each last digit (the tenths place) to the right of the vertical line in the row corresponding to its leading digits. For instance, the first three mileages—30.8, 31.7, and 30.1—are arranged as follows:

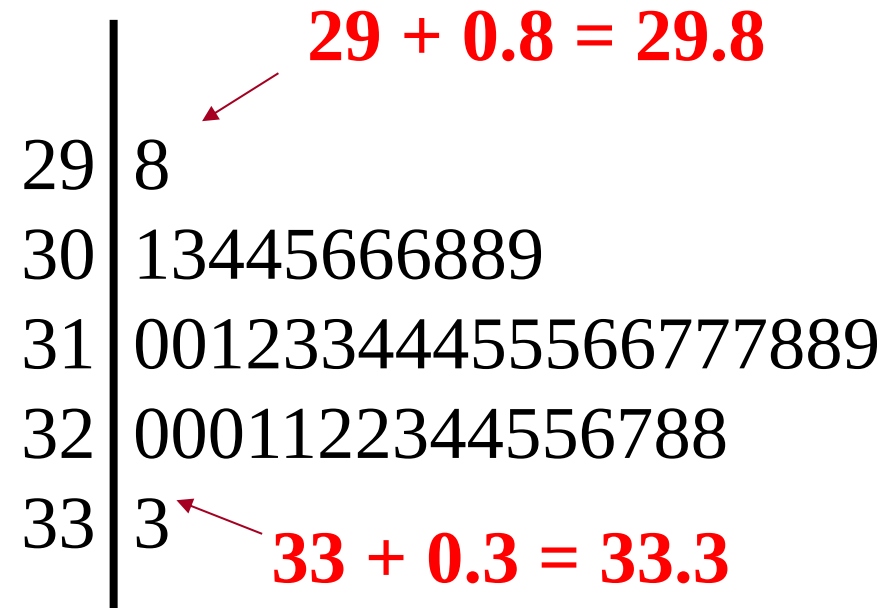
29		
30		8 1
31		7
32		
33		

We form the leaves of the display by continuing this procedure as we pass through all 50 mileages. After recording the last digit for each of the mileages, we sort the digits in each row from smallest to largest and obtain the stem-and-leaf display that follows:

29		8
30		1 3 4 5 5 6 7 7 8 8 8
31		0 0 1 2 3 3 4 4 4 4 4 5 5 6 6 7 7 7 8 8 9 9
32		0 1 1 1 2 3 3 4 4 5 5 7 7 8
33		0 3

The numbers to the left of the vertical line form the stem of the display. Each number to the right of the vertical line is a leaf. Each combination of a stem value and a leaf value represents a measurement in the data set.

The stem-and-leaf display of car mileages:



Questions (at the end of the session and **if time permits**)

2.13 Explain:

- a) Why we construct a frequency distribution and a histogram for a data set.
- b) The difference between a frequency histogram and a frequency ogive.

2.14 Explain how to find:

- a) The frequency for a class.
- b) The relative frequency for a class.
- c) The percent frequency for a class.

2.16 Consider the following data:  HistoData

36	39	36	35	36	20	19
46	40	42	34	41	36	42
40	38	33	37	22	33	28
38	38	34	37	17	25	38

- a** Find the number of classes needed to construct a histogram.
- b** Find the class length.
- c** Define nonoverlapping classes for a frequency distribution.
- d** Tally the number of values in each class and develop a frequency distribution.
- e** Draw a histogram for these data.
- f** Develop a percent frequency distribution.

The largest data is 46, the smallest data is 17

Step 1: Number of classes

- For $K = 4$, $2^4 = 16$, < 28
- For $K = 5$, $2^5 = 32$, > 28
- So use $K = 5$ classes

Step 2: Class length

- For $(46 - 17) / 5 = 5.8$
 - round **up** to 6

Step 3: Form non-overlapping classes of equal width

Class 1	17 and less than 23
Class 2	23 and less than 29
Class 3	29 and less than 35
Class 4	35 and less than 41
Class 5	41 and less than 47

Step 4: Tally and count the number in each class

Class	Tally marks	Frequency	Relative frequency
17 < 23	/ / / /	4	0.143
23 < 29	/ /	2	0.0714
29 < 35	/ / / /	4	0.143
35 < 41	/ / / / / / / / / / / / / /	14	0.5
41 < 47	/ / / /	4	.143

A quick re-cap

- Frequency distribution
- Relative frequency, percent frequency
- Cumulative distribution (ogive)
- Skewed Distributions
 - Dot Plots
 - Stem-and-Leaf Displays

Thank you!